Hybrid Artificial Neural Networks for Modeling Shallow-Water Bathymetry via Satellite Imagery

Mosbeh R. Kaloop, Mohammed El-Diasty, Jong Wan Hu, and Fawzi Zarzoura

***Abstract*— Accurate bathymetric mapping for shallow-water areas is essential for coastal and maritime engineering applications. However, traditional multibeam or light detection and ranging (LiDAR) survey techniques used to produce high-quality bathymetric maps are expensive. Satellite-derived bathymetry provides a fast and inexpensive method for the large-scale mapping of shallow-water areas and can overcome the complexities of traditional bathymetric mapping methods in these areas. Traditionally, linear regression models, most commonly the Stumpf model, are used for satellite-based bathymetric modeling. However, nonlinear artificial neural network (ANN) models have been recently developed and implemented for satellite-based bathymetric modeling and are under significant investigation to develop the most accurate and optimal model. This article proposes two new hybrid ANN-based models for bathymetric modeling and investigates their performance using satellite imagery data and "truth" depth data for a coastal shallow-water study area. Two-hybrid ANN algorithms are developed, namely, particle swarm optimization (PSO)-ANN and optimally pruned extreme learning machine (OPELM), and their results are compared with the traditional Stumpf method and current state-of-the-art ANN model. The study area dataset comprises the "truth" depth data from a nautical chart of the Alqumriyah Island study area in Saudi Arabia and the corresponding spectral reflection values of green, blue, and near-infrared bands from the free-of-charge Level-1C product of Sentinel-2A images used to train and validate the two newly developed models and the traditional models. The results show that the developed OPELM method can accurately derive the bathymetry and is superior to the developed PSO-ANN model, the current state-of-the-art ANN model, and the traditional Stumpf model by 12.10%, 18.76%, and 32.46%, respectively. The OPELM model can also be used for bathymetric modeling of shallow-water areas with depths up to 30 m with a high level of accuracy compared with the current state-of-the-art ANN and traditional methods. The significant contribution of this research is that it is the first investigation of the artificial intelligence-based hybrid OPELM method for accurate bathymetric modeling and will certainly encourage further investigations of hybrid models. Moreover, this research explores whether these developed hybrid models can meet the International Hydrographic Organization standards for hydrographic survey applications.**

***Index Terms*— Artificial neural network (ANN), bathymetric, International Hydrographic Organization (IHO), modeling, Satellite-2A.**

I. Introduction

SHALLOW-WATER bathymetry is an indicator of the characteristics of lakes, seas, or coastal ocean floors; thus, accurate modeling or mapping is essential for coastal and maritime engineering applications and shipping safety. Nowadays, light detection and ranging (LiDAR) operated from aerial platforms and autonomous underwater vehicles has been used effectively to determine coastal water depth [1]. However, to overcome the expenses of hydrographic surveying for bathymetric modeling, free-of-charge Landsat satellite images are utilized extensively [2]–[5]. Other satellite images have also been used, such as Sentinel-1 and -2, SPOT, IKONOS, and WorldView (WV) [4], [6]–[11]. The main advantage of satellite tools for bathymetric mapping is that multispectral data can be collected at different times and locations, and accurate bathymetric maps can be extracted from these data based on seafloor reflection analyses of satellite bands in the water column [4], [12], [13]. As the accuracy of satellite seabed mapping depends on water clarity, this approach has been applied typically over areas with clear water [14]. In shallow-water regions, water clarity has a significant impact on bathymetric mapping accuracy using satellite imagery [1], [14]. In addition, many researchers have studied different applications to improve satellite images over clear water [1], [14]–[16]. Shallow-water bathymetry extracted from Sentinel-2 images is proposed in this study. Sentinel-2 can be used for accurate bathymetric modeling compared with Landsat 8 and high-spectral-resolution images, such as WV2 [16]. Sentinel-2 data are a feasible correlative source of preexisting moderate-resolution images, with an expanded spectral resolution, and achieves continuity with SPOT and Landsat-type image data [17].

Sentinel-2 possesses 13 spectrum bands [ranging from visible and near-infrared (NIR) to shortwave infrared wavelength (SWIR)]: four bands at 10 m, which include blue (B), green (G), red (R), and NIR; six bands at 20 m, which include four narrow bands in the R-edge spectral domain

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and two large SWIR bands; and three bands at 60 m, which are dedicated mainly to atmospheric corrections and cloud screening [3], [15], [17], [20]. The accuracy of the image size can be improved by resampling visible and NIR bands, whereas NIR and SWIR are used for atmospheric corrections [21]. Many researchers have developed the use of different spectral bands to classify surface water areas using satellite images [4], [12], [13], [17], [19], [20]. NIR and/or SWIR bands are generally used for water area detection, whereas most radiations of NIR wavelengths and beyond are absorbed by other landscape features. The R, NIR, and SWIR bands have been used to classify land cover, with promising results [22]. The R and NIR spectral bands are not beneficial for most water areas, but land areas can be extracted using the NIR band only [17], [23], [24].

Meanwhile, bathymetric modeling can be achieved based on the relationship between satellite imagery spectral bands and water depth [13]. Bathymetric modeling using R, G, and B bands was traditionally developed from linear model functions [9], [10], [25], although both linear and nonlinear features are embedded in satellite images and can be used to improve models. Traditionally, linear models, such as the Stumpf method, were applied as a simple bathymetric model [9]. However, the accuracy of linear bathymetric models is limited, and potential nonlinear models are under investigation to develop more accurate bathymetric models [4].

Nonlinear models have been recently applied to improve bathymetric modeling based on satellite imagery. For instance, Mohamed *et al.* [15] used support vector regression to model lake bathymetry, and their results showed that the accuracy of the proposed model is higher than that of linear models. Vojinovic *et al.* [26] applied the support vector machine (SVM) to model shallow-water depth based on satellite image data; the proposed model was accurate for depths less than 15 m with a mean absolute error (MAE) = 1.8 m. Artificial neural network (ANN) bathymetry models were found to be powerful nonlinear modeling tools [27]. The current and previous state-of-the-art applications of ANN for bathymetric modeling based on satellite imagery were proposed previously [2], [27]. For example, a lake shoreline bathymetry map was detected by an ANN, and the results showed high accuracy in determining the lake water depth from satellite image data [5]. Ceyhun and Yalçin [12] applied an ANN to map the water depth up to 40–45 m, and showed that it was efficient in estimating water bathymetry. The root mean square errors (RMSEs) of ANN models for water depth detection are generalized as 0.36–4.36 m at depths of 15–18 m [27]. More details about using soft computing techniques to estimate the water depth can be found in previous studies [28]–[31].

More recently, hybrid algorithms have been investigated for nonlinear modeling; for example, hybrid bio-optical transformation improved bathymetry obtained from satellite images [32]. Furthermore, nonlinear hybrid ANN models are more powerful for improving the accuracy of ANN models of water depth and level and ocean wave height [33]. Different algorithms are integrated with ANNs to improve models in engineering applications [34], [35]. Wavelet and particle swarm optimization (PSO) improved ANN modeling results [4], [36], [37]. Wavelet ANNs (WNNs) improved the accuracy of ANNs for bathymetric modeling [4]. PSO automatically adjusted ANN parameters [38]. PSO-ANN enhanced ANN models in different engineering applications, such as water level and groundwater detection and other applications [37]–[39], but is still limited for use in bathymetric modeling. In addition, the optimally pruned extreme learning machine (OPELM) is still a new hybrid algorithm based on ANN theory [40], and is not used for modeling bathymetry. Previous studies showed that extreme learning machine (ELM) models outperform ANN models in different applications [37], [41], [42]. In OPELM, the neuron numbers in hidden layers can be determined to overcome the classification and regression problems of ANN modeling [43].

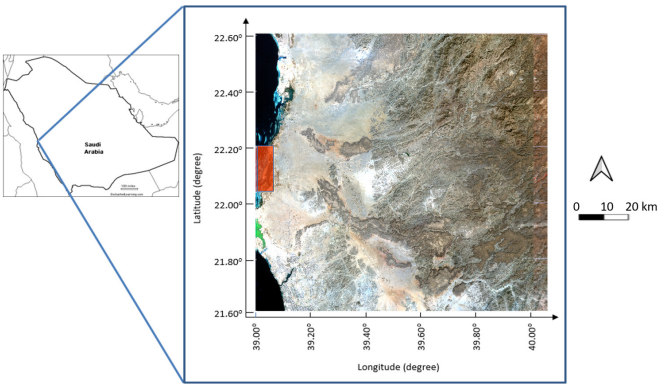


Fig. 1. RGB image for the study area (R box).

In this research, a free-of-charge Level-1C multispectral satellite image from the Sentinel-2A multispectral instrument (MSI) mission was used to design and develop the proposed PSO-ANN and OPELM models. To validate the developed models, a comparative study was conducted between the developed nonlinear hybrid PSO-ANN and OPELM models, the most common traditional Stumpf log-ratio method, and the current state-of-the-art ANN model. This article investigates the accuracy of promising highly nonlinear hybrid neural network models for satellite-derived bathymetric modeling based on a limited number of sparse depth data points and check whether the accuracy of the developed models can meet the International Hydrographic Organization (IHO) standards for the hydrographic surveying of different survey orders [4], [44].

II. Material and Methods

A. Study Area and Data Sources

Alqumriyah island study area, Saudi Arabia, along the Red Sea, was chosen to investigate the performance of the proposed hybrid neural network models (as seen in Fig. 1) and explore the effectiveness of satellite-2A images for bathymetric modeling of shallow water areas with a limited number of truth depth data. The selected study area is considered a typical shallow water area to investigate where the depth ranges are from 0 to 30 m and contain different onshore and offshore intertidal areas. The available truth depth data measurements were obtained from the nautical chart developed from collected hydrographic data using multibeam

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TABLE I
WATER DEPTH STATISTICAL ANALYSIS

Depth	Points No.	Mx	Mn	Av	STD
0-5	41	4.80	0.50	2.74	1.44
5-10	22	9.80	5.10	7.04	1.48
10-15	15	14.90	10.00	12.01	1.41
15-20	6	17.70	15.70	16.47	0.96
20-30	6	28.00	20.00	24.67	2.62

echo-sounding system owned by the General Commission of Surveys (GCS) of Saudi Arabia. The "truth" depth data points (90 points) were extracted from the nautical chart created in December 2013, where it is not possible to access the hydrographic data collected by GCS. Table I presents statistical analysis [maximum (Mx), minimum (Mn), average (Av), and standard deviation (STD)] of truth depths' measurements. From Table I, it can be seen that the distribution of collected data was shown high at the shallow water up to 15 m, 78 points; while 12 points are available for depth between 15 and 30 m. In addition, statistically the estimated depths within the ranges 0–5, 5–10, and 10–15 are almost normally distributed, while the remaining estimated depths are not normally distributed. Although the data available from GCS are not distributed well over the study area, the satellite image can be employed to estimate empirical water depth to optimal management [45], [46].

B. Data Sources and Processing

In this article, the Level-1C satellite image product of the Sentinel-2A mission operated by the European Space Agency (ESA) was used to extract the data for bathymetric evaluations and analyses. The Level-1C satellite image product was acquired in November 2016 for the study area of Alqumriyah Island, Saudi Arabia, along the Red Sea. Bathymetric modeling using the high-resolution Sentinel-2 imagery outperforms bathymetric modeling accuracy using low-resolution Landsat 8 imagery because of the image resolution [20], [46]. The temporal resolution of Sentinal-2A is greater than that of Landsat 8 at the equator; also, the swath width of Sentinel-2 is 290 km as compared with 185 km of Landsat 8 [20]. The time difference between image acquisition and chart creation is about three years; although this is considered as a short time period, it can make a considerable change in bathymetry, as bathymetry needs decades to change significantly.

In this article, the Level-1C top-of-atmosphere (TOA) spectral reflectance values for the multispectral bands were used to estimate the Level-2A product bottom-of-atmosphere (BOA) spectral reflectance values, where Level-2A products were not available for the study area. Then, the Level-2A product BOA reflectance values for the multispectral bands were utilized to construct the proposed models. The accuracy and characteristics of the commonly used spectral bands are presented in Table II. Sentinal-2A, with 12-cycle digitization and B2 band at 490 nm, gives a high resolution of penetration in clear (open) waters (Table II) and is useful for strategies dependent on radiative transfer models. The spatial resolution

TABLE II
BAND SPECIFICATIONS OF SENTINEL-2A SPECTRAL BAND SETTINGS USED IN THIS STUDY INDICATING RANGE WAVELENGTH, WIDTH, SPATIAL RESOLUTION, Δt, AND ASSOCIATED SIGNAL TO NOISE RATIO (SNR) AT REFERENCE RADIANCE [47], [48]

Band	Wavelength range (nm)	Special resolution (m)	Bandwidth (nm)	SNR	Δt (Sec)
B2-Blue	448–546	10	66	154	0
B8-NIR	763–908	10	106	174	0.264
B3-Green	538–583	10	36	168	0.527
B4-Red	646–684	10	31	142	1.005
B8A-NIR	848–881	20	21	72	2.055

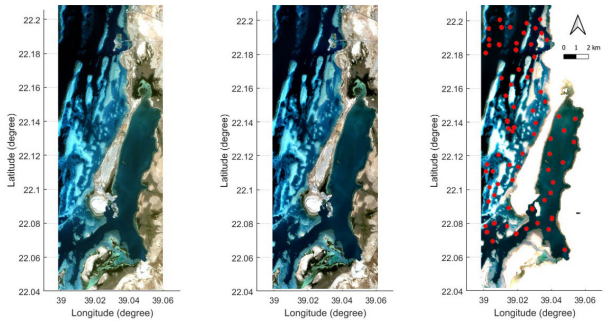


Fig. 2. (Left) Study area location and satellite images processing: RGB image of Level-1C. (Middle) RGB image of Level-2A. (Right) Masked and filtered RGB image of Level-2A generated along with selected stations, R circles.

of 10 m in four visible and NIR bands lies regimentally between the ∼2-m pixel high-resolution sensors (see Table II and [16]). The time lag (Δt) between the detector bands and the first band (B2 band) increases from 0.264 to 2.055 s for the NIR B8 and B8A bands; however, the time lag of the R band shows a higher value in Table II. The time lag of 10-m resolution bands ranges up to 1.005 s, making the R–G–B (RGB) bands useful in the phase shift for bathymetry mapping. In addition, from Table II, it can be concluded that there is good spectral correlation for water objects in the B and G spectral range. Fig. 2 presents the RGB image of Level-1C created from the Sentinel-2A satellite; the image size is 100 × 100 km for the study area.

Single (coastal (C), B, and G) and ratio (C/B, B/C, B/G, and G/B) bands of satellite images were investigated and studied, and the results show that the ratio bands are better for bathymetric mapping [49]. In addition, Muzirafuti *et al.* [7] summarized that the BG bands should be commonly used, because they can penetrate up to 30 m in shallow clear water. Also, high-precision bathymetry mapping was determined using RG bands [49]. In this article, multiple spectral reflectance bands, G (R_G), B (R_B), and NIR, were employed. Fig. 2(left) presents the RGB image of TOA spectral reflectance values for the Level-1C product of Sentinel-2A multispectral bands. To improve image accuracy, the systematic image errors are adjusted and noises are approximately reduced [50]. The topography, cirrus, and atmospheric effects were adjusted in the preprocessing step using sentinel application platform (SNAP) software [51] with sen2cor toolbox [52] to provide BOA spectral reflectance values

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for the Level-2A product of Sentinel-2 multispectral bands. The atmospheric correction with sen2cor toolbox is performed using a set of look-up tables generated via libRadtran [52]. The employed sen2cor toolbox processing configuration parameters were the maritime option for aerosol correction, yes option for digital elevation model correction for topographic correction, mid-latitude winter option for atmosphere profile, and yes option for cirrus correction. Fig. 2(middle) shows the RGB image of BOA spectral reflectance values derived from the sen2cor toolbox. In addition, the normalized difference water index (NDWI) ratio technique using a 0.05 threshold limit was applied to mask out the nonwater regions in preprocessing. The NDWI [4] can reduce the effect of nonwater classes in the satellite image using the NIR band and the G band (NIR − G/NIR + G). Notably, to enhance the image quality in preprocessing, a spatial domain median filter with a 3 × 3 kernel size window was applied to remove embedded noise in the image. Fig. 2(right) shows the masked and filtered RGB image of the Level-2A product along with designated stations with known chart depths.

To develop the proposed models, the extracted data of 90 collocated stations, shown in Fig. 2(right), were used; these stations are visualized on the masked and filtered RGB image of the Level-2A product. These points also have known BOA spectral reflectance values of the Level-2A product and corresponding marine chart depths. In the modeling and validation steps, the PSO-ANN, OPELM, ANN, and Stumpf models were developed and evaluated using the BOA spectral reflectance values of the G and B bands. Consequently, the proposed hybrid neural network models, PSO-ANN, and OPELM were compared with the traditional ANN and Stumpf log-ratio models and discussed. Finally, the novel hybrid ANN model accuracies were examined to check whether they can achieve IHO standards for hydrographic survey requirements of different survey orders [4], [44].

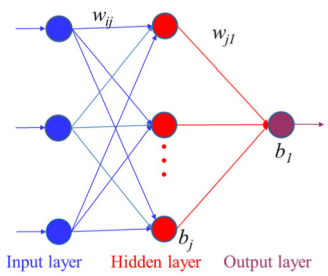


Fig. 3. Single hidden layer feedforward ANN structure.

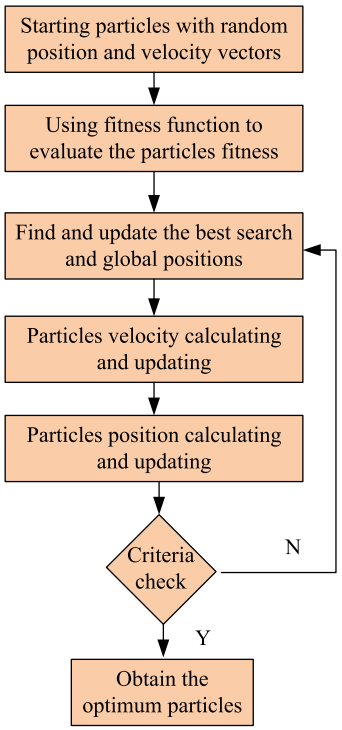


Fig. 4. PSO flowchart [59].

C. Model Design and Evaluation

Two-hybrid models were evaluated and assessed in this study; PSO-ANN and OPELM models were developed for shallow-water bathymetric modeling. One hidden layer feedforward ANN was used in all models because the design of the ELM is the same.

1) PSO-ANN: The PSO-ANN is an integrated system in-between PSO and ANN. In this study, the PSO is used to enhance ANN parameters, variables of weights (*wij* and *wj*1), and biases (*bj* and *b*1). PSO details can be found in [53], [54]. Depth state-of-the-art of ANN design, theory, and applications can be found in [55]–[57]. A summary of both algorithms and integrated techniques is proposed. In general, the ANN comprises three layers, input, hidden, and output; each layer includes one or more neurons [56], [58]. In this research, the neuron numbers in the input layers were three inputs. The neuron numbers in the hidden layers are traditionally determined by trial and associated error method; in this, five neurons were found optimal for hidden layer neurons. The relative accuracy of this model according to the overall STD is 11.11%. To verify the proposed structure, 3-, 10-, 15-, and 20-neurons were evaluated and the relative accuracies were 15.70%, 12.08%, 16.23%, and 24.26%, respectively. The hidden layer neurons are utilized to link the input and output layers by a nonlinear or linear transfer function. The Levenberg–Marquardt function is employed for training the ANN model. The output layer neurons consist of the number of outputs; in this, the water depth was estimated. Fig. 3 illustrates a typical ANN model.

The PSO is an algorithm that utilizes particles that move in the group for disclosing better results. A vector of those particles, called velocity vector, is utilized to update the position of particles. The standard PSO flowchart is presented in Fig. 4. The objective of the swarm is calculated by an objective function, which is used for fitness of each particle determining [59]. After that, the velocity of the particles is defined using a function that depends on the best positions of the swarm and each particle based on (1). Then, the following position of all particles is determined through their velocity using (2). The process by iteration and specific accuracy can be applied until the consequence criterion of the swarm

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is identified

$$v_{\text{new}} = wv + r1.C1.(P\text{best} - P) + r2.C2.(G\text{best} - P) \quad (1)$$

$$P_{\text{new}} = P + v_{\text{new}} \quad (2)$$

where w is an inertia weight; v and v_{new} represent the current and updating velocity vectors of particles, respectively; P and P_{new} denote the current and updating position vectors of particles, respectively. $C1$ and $C2$ are predefined coefficients; Pbest and Gbest are the best position of the particle in the research stage, and the global best position among all particles, respectively. $r1$ and $r2$ refer to a constant interval in-between 0 and 1 sets randomly.

In the integrated PSO-ANN algorithm, the process begins with creating the initialization of a group of random particles where the positions of these particles randomly represent the weights and biases of the ANN. Afterward, the PSO-ANN is trained using the initial weights and biases (i.e., the initial positions of particles) and the model error is calculated. Consequently, the model error is decreased with changing the particle positions at each iteration. Then, the Pbest (i.e., the lowest error by each particle at that moment) and Gbest (i.e., the lowest error by all particles at that moment) are used to update the velocity equation [see (1)]. As a result, the output value from (1) is produced to adjust particle positions to the best solutions. In the end, a new error is obtained using the updated positions. This procedure continues until the termination criteria are met, with the optimization output obtained for the value of Gbest. The following steps summarize the proposed algorithm [59]; 1) start with ANN train and setting ANN parameters (weights, biases, and number of hidden neurons), and specify mean square error (MSE) as a fitness function; 2) initialize PSO particles positions (P) and train the ANN; 3) estimate the model error and learning error by setting overall error as Gbest and each particle best error as Pbest; 4) calculate velocity and update position based on Gbest and Pbest particle; 5) train ANN using new particle position; and 6) repeat steps 3, 4, and 5 until meet with learning target error.

2) OPELM: OPELM was introduced by Miche *et al.* [60]. It is designed based on ELM to overcome its problems and make it more robust and generic [60]. The OPELM outperforms SVM, multilayer perceptron (MLP), Gaussian process (GP), and multilayer perceptron neural network (MLPNN) in the nonlinear modeling [60], [61]. The structure of OPELM is a single-layer feedforward ANN model, as the same in Fig. 3, with a fast selection of the weights [60]. The main advantage of ELM or OPELM is it can optimize the ANN parameters, including neurons numbers of the hidden layer. The algorithm of ELM can be presented as follows.

The output of the ELM mainly depends on the matrix H, and this matrix can be solved by gradient-based algorithm optimization; otherwise, the kernel function is used to solve the ELM [62]. The ELM selects the kernel function randomly; thus, the OPELM is developed to select the kernel function robustly. The steps of OPELM can be summarized as follows [60], [63] (see Fig. 5): 1) using ELM to design the ANN structure with a lot of neurons; 2) using multiresponse sparse regress (MRSR) to classify ranking of the best neurons that can improve the model output; in this study, the weight matrix is used to make this classification; and 3) select the optimum neurons for the modeling using the leave-one-out (LOO) validation method. In OPELM, three kernel functions, linear, sigmoid, and Gaussian kernels, are used, and the OPELM can select the best one for the modeling.

Algorithm 1 ELM [60]

Consider a training set (x_i, y_i), $x_i \in R^n$, $y_i \in R^m$ (I = 1, 2,…, M), where n and m are the number of observations of input and output variables, respectively; an activation function f: $R \to R$, and N hidden neurons:
1: Randomly assign input weight w_i and biases b_i, $1 \le i \le N$.
2: Determine the hidden-layer output matrix H.
3: Calculate output weights matrix $\beta = H^*Y$, where H^* is the Moore-Penrose generalized inverse of H, and Y refers to the optimum values of ELM.

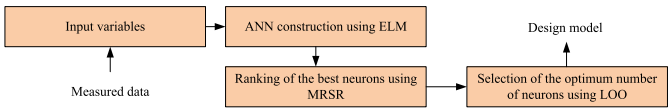


Fig. 5. OPELM processing steps [60].

3) Model Processing and Evaluation: In this article, the proposed models were compared with the traditional Stumpf log-ratio method (Stumpf method) [4], [9], and the normal ANN model for bathymetric modeling. All models were built and processed using MATLAB software [64]. The bathymetry (d) can be estimated using the Stumpf method as follows [4], [9]:

$$d = m_0 + m_1 \frac{\ln(R_{\text{G}})}{\ln(R_{\text{B}})} \quad (3)$$

where m_0 and m_1 are the unknown parameters; and R_{G} and R_{B} are the BOA reflectances of the G and B bands, respectively. It should be mentioned that, according to [4], [7], [9], [49], and (3), the log ratios of the G and B bands provide high-accuracy linear bathymetric modeling. This fact was also investigated by estimating the correlation of the log ratios of RGB bands with actual depth values; the results showed that the correlation coefficient of ln(B)/ln(G) with real bathymetry is greater than those of ln(G)/ln(R) and ln(B)/ln(R) with real bathymetry [6]. Therefore, the log ratio of B and G bands was used in this study.

All three ANN models were designed with three input neurons and one output neuron. The input neurons are the log of G and B bands and the log ratio of both bands, as in (3), whereas the output is the "truth" nautical chart depth. The traditional ANN model was designed using different ANN architectures for the number of neurons and weights and biases matrices. The ANN model structure was determined based on the lowest model error, and the optimum structure for bathymetric modeling in this study was found to be seven hidden neurons with a nonlinear hyperbolic tangent (Tan$h$) activation function for a model structure of three inputs, seven hidden neurons, and one output (3-7-1).

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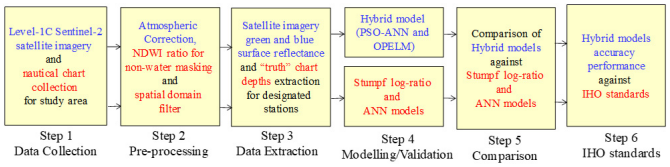


Fig. 6. Models processing and evaluation.

In the PSO-ANN model design, five neurons were chosen for the hidden layer after trial and error. Thus, the model structure was (3-5-1) with the PSO solution. Initially, random weights and hidden biases were selected, and the output weight and biases were calculated using the basic ANN algorithm. The PSO was used to obtain the optimal weight (3×5) and bias ($5 \times$ m) matrices based on 1000 iterations and the Mn MSE. In this study, 20 decision variables at each iteration for the PSO population were calculated from the ANN learning parameters. These parameters were augmented through the training stage to attain the best input weights and hidden biases. PSO is affected by its essential parameters, such as the number of populations, acceleration coefficients, and inertia weight. After some opening trials, the swarm dimension was considered as 10, $C1 = 1.5$, $C2 = 2.5$, and inertia weight $= 0.185$.

In OPELM, as in PSO-ANN, random input matrices of weights and biases were assumed, and the ELM algorithm was used to estimate the output weights and bias. The ELM was used to extract the optimal final weights and bias matrix connections of input and output neurons. The ELM can select the number of neurons; so, 50 neurons were designed as the Mx neurons that can be generated based on the number of training datasets. Also, the LOO error was used to determine the optimal number of hidden neurons. Furthermore, the optimum sigmoid kernel function was defined and extracted by OPELM to build the optimal model. The final structure of OPELM for bathymetric modeling was found to be (3-15-1).

Fig. 6 illustrates the six stages of model implementation: 1) generate data through satellite images and truth depth data; 2) improve the satellite image datasets and filter the extracting datasets; 3) design the data for modeling based on the available truth data; 4) develop and validate the proposed models; 5) compare the proposed models and select the optimal model for bathymetric modeling of shallow-water areas; and 6) compare the model with IHO standards to check whether the accuracy achieved meets the IHO standards for hydrographic survey requirements of different survey orders [4], [44]. The total data of 90 points were divided randomly into training (70%) and testing (30%) stages. The training set was used to design the models, and the testing set and all the data were used to examine the proposed models. Table III documents the statistical analyses of input and output data variables for the training and testing phases.

It can be seen from Table III that the sample depth data distributions of the training and testing points covered the whole study area. The training and testing datasets covered a wide range of water depths in the study area, to 27.4 and 25.0 m, respectively, which increases confidence in the developed models.

TABLE III
STATISTICAL ANALYSIS OF TRAINING AND TESTING DATASETS

STATISTICAL PARAMETER	LN(R_G)	LN(R_B)	LN(R_G)/LN(R_B)	TRUTH
TRAINING				
MX	-1.98	-1.94	1.12	28.00
MN	-4.23	-5.47	0.72	0.60
AV	-3.10	-3.53	0.90	8.12
STD	0.62	1.03	0.09	6.55
TESTING				
MX	-1.98	-1.98	1.13	25.50
MN	-3.96	-5.24	0.76	0.50
AV	-3.08	-3.37	0.93	6.77
STD	0.58	0.92	0.08	6.15

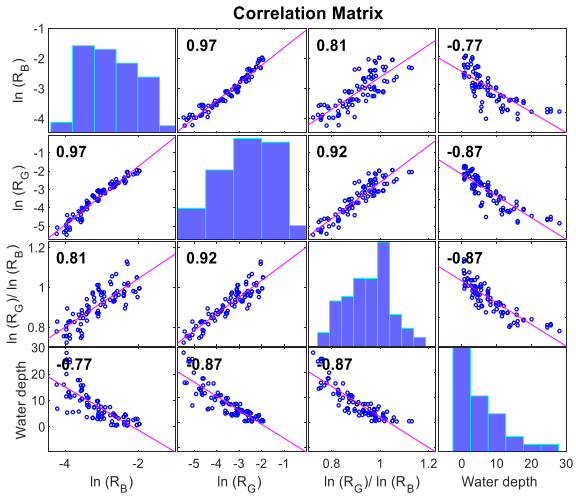


Fig. 7. Correlation matrix of inputs and output variables.

In this study, three statistical parameters are used to evaluate the performance of the proposed models, Pearson's correlation coefficient (R) (4), RMSE (5), and MAE (6). In addition, the violin plot was extracted to assess the model's accuracy and understand the model's behaviors

$$R = \frac{\mathrm{COV}(d_o, d_p)}{\sigma_{d_o}\sigma_{d_p}} \times 100 \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{M}(d_o - d_p)^2}{M}} \tag{5}$$

$$\mathrm{MAE} = \frac{1}{M}\sum_{i=1}^{M}|d_o - d_p| \tag{6}$$

where $\mathrm{COV}(d_o, d_p)$ is the covariance between nautical chart depths (d_o) and estimated depth (d_p); σ_{d_o} and σ_{d_p} are the STDs of the nautical chart depth and estimated depth, respectively. M is the number of dataset.

III. RESULTS AND DISCUSSION

First, the effects of extracted bands of satellite images on water depth should be understood. Thus, the correlations between input (satellite image bands) and output (chart depths) variables were computed, as presented in Fig. 7. A high negative correlation is clearly shown in Fig. 7 between the three input variables (B, G, and band log ratio) and chart depth.

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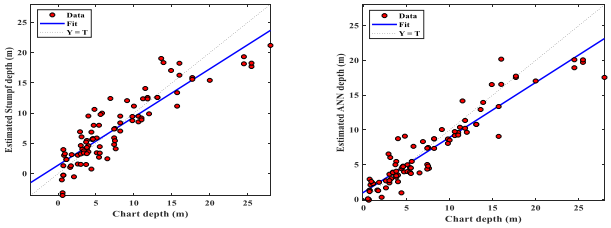


Fig. 8. Scatter plot of Stumpf log-ratio and ANN models results with chart depth.

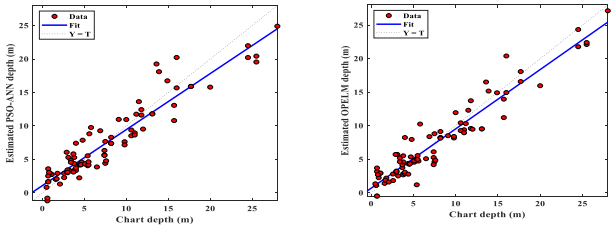


Fig. 9. Scatter plot of PSO-ANN and OPELM models depth with chart depth.

The G band has a higher impact than the B band on bathymetric modeling, and the band log ratio also influences the water depth modeling, corresponding with Stumpf *et al.*'s results [9]. In addition, the distribution of input and output variables has a nonlinear relationship.

For further investigation, three OPELM models were evaluated to assess the performance of input bands of satellite images on model accuracy. The network structure of Model 1 was designed to receive two inputs, which are the B band and the log ratio of the two bands. The network structure of Model 2 was designed to receive two inputs, which are the G band and the log ratio of the two bands. The network structure of Model 3 was designed to receive three inputs, which are the B and G bands and the log ratio of the two bands. The model training data were used to estimate the correlation of each estimated depth in the models with chart depths, and the correlations of models 1, 2, and 3 were 95.9%, 94.9%, and 96.4%, respectively. These results indicate that the three input variables of satellite image bands affect bathymetric model performance.

The proposed models to estimate bathymetry through satellite imagery were evaluated and are presented in Figs. 8 and 9 and Table IV for the training, testing, and overall (all) datasets. Fig. 8 illustrates the performances of the Stumpf and ANN models for overall data. It can be seen from Fig. 8 that the linear fitting slopes are 0.78 and 0.79 for the Stumpf and ANN models, respectively. In addition, from Table IV, it can be seen that the ANN model outperforms the Stumpf model in all data phases. The overall RMSE values of the ANN and Stumpf models are 2.36 and 2.84 m, respectively, and the correlation of estimated depths by the ANN model is obviously improved compared to the Stumpf model. This is because of the high nonlinearity of the ANN model.

Fig. 9 presents the satellite-estimated bathymetry using PSO-ANN and OPELM models. From Fig. 9 and Table IV, it can be seen that bathymetry estimated by the PSO-ANN model is improved by 7.58% and 23.18% compared with the ANN and Stumpf methods, respectively, in terms of RMSE. In addition, the linear fitting slope between the estimated and chart depths is 0.83 for the PSO-ANN. The overall correlation of PSO-ANN is increased by 4.5% and 0.3% compared with the ANN and Stumpf methods. From these results and comparisons of PSO-ANN with ANN and Stumpf methods in the training, testing, and overall stages, it can be concluded that PSO improved the performance of ANNs for bathymetric modeling.

TABLE IV
PERFORMANCE ASSESSMENT AND COMPARISON OF PROPOSED MODELS

Model	Data phase	MAE (m)	RMSE (m)	R (%)
Stumpf log-ratio	Training	2.08	2.78	90.40
	Testing	2.45	2.96	87.88
	All	2.19	2.84	89.68
ANN	Training	1.43	2.26	94.46
	Testing	2.00	2.57	92.87
	All	1.60	2.36	93.94
PSO-ANN	Training	1.63	2.12	94.56
	Testing	1.85	2.31	94.36
	All	1.70	2.18	94.21
OPELM	Training	1.33	1.73	96.41
	Testing	1.81	2.29	94.57
	All	1.48	1.91	95.51

Fig. 9 shows the scatter plot of the estimated water depth by the OPELM model with chart depths; from Fig. 9 and Table IV, it can be seen that the linear fitting slope is higher than that in other models, at 0.88. In addition, the performance of satellite-estimated bathymetry using OPELM is higher than the other models in training, testing, and overall evaluation phases, as presented in Table IV. The overall accuracy of OPELM is improved in terms of the MAE by 32.79%, 7.73%, and 13% compared with Stumpf, ANN, and PSO-ANN, respectively. Furthermore, the overall accuracy of OPELM is improved in terms of RMSE by 32.46%, 18.76%, and 12.10% compared with Stumpf, ANN, and PSO-ANN, respectively. The correlations between the model results and nautical chart depths are 96.41%, 94.57%, and 95.51% for the training, testing, and overall stages, respectively. These results reveal that satellite-estimated bathymetry using OPELM is the best model among the four models studied for use in satellite-derived bathymetric mapping.

The errors of the four models were further evaluated and assessed. Fig. 10 shows the overall model errors. The Mx errors are 7.79, 10.44, 5.94, and 4.48 m for satellite-estimated bathymetry using Stumpf, ANN, PSO-ANN, and OPELM models, respectively. The error ranges using Stumpf, ANN, PSO-ANN, and OPELM models were 13.62, 15.18, 11.61, and 8.94 m, respectively.

A violin plot of the model errors is shown in Fig. 11 to visualize bathymetric model behaviors. The violin plot presents the model error distributions and peaks in the errors. The distribution of OPELM model errors for satellite-estimated bathymetry is approximately normal, with an equal mean

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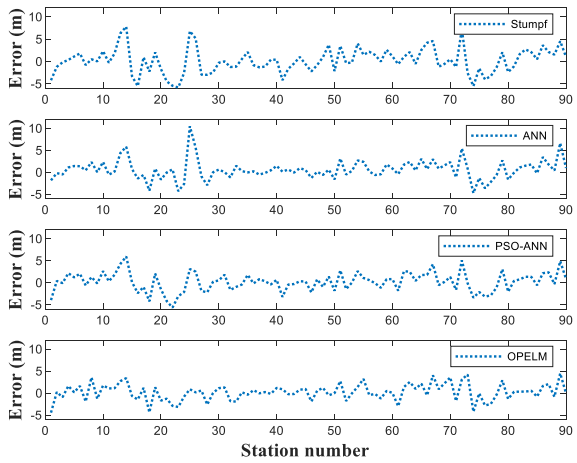


Fig. 10. Models errors for the whole data.

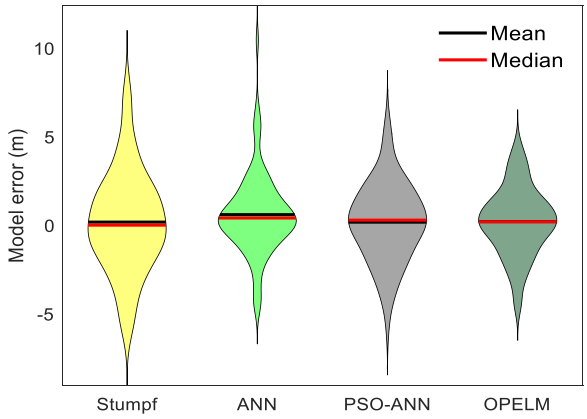


Fig. 11. Violin plot of model's errors.

and median of the model error; in addition, the distortion of the model is obviously lower than that of the other models. This indicates the high accuracy of the OPELM model for bathymetric modeling based on satellite image data.

Model accuracies and performances were investigated based on the RMSE values, and statistical parameters of the model errors to estimate the water depth at different depth ranges are presented in Table V. Model performance is depth-dependent, and accuracy changes according to the depth range. However, it should be noted that the number of data points in the depth ranges given in Table V cannot be considered equally weighted by meter due to the limited number of data points used in the study area under investigation that is not equally distributed over the respected depth ranges. From Table V, PSO-ANN outperforms the other models in the range 0–5 m. In comparison, the ANN model shows better performance for the depth range of 5–15 m. The OPELM model provides accurate depth estimation in the range of 15–30 m. Also, the OPELM model has good performance at depths that possess a small number of "truth" depth data points. This implies that the OPELM model can be considered the best model among the other models proposed for shallow-water depth estimation when a limited number of truth depth data points are available. In addition, the overall performance of the proposed models is presented in Tables IV and V and Figs. 8–11. From these results, it can be concluded that

TABLE V
MODELS PERFORMANCES AT A DIFFERENT WATER DEPTH

Depth	Statistical	PSO-NN	OPELM	Stumpf	ANN
0-5	Mx	2.15	1.61	4.20	3.45
	Mn	-3.42	-4.25	-5.84	-4.74
	Av	-0.64	-0.60	-0.59	-0.42
	RMSE	1.48	1.50	2.27	1.59
5-10	Mx	3.07	4.19	4.03	3.12
	Mn	-3.97	-4.46	-5.53	-1.84
	Av	0.57	0.71	0.08	0.86
	RMSE	1.97	1.94	2.55	1.64
10-15	Mx	2.46	3.57	2.10	2.34
	Mn	-5.67	-2.92	-5.45	-2.67
	Av	-0.21	0.60	-0.55	0.86
	RMSE	2.37	2.07	2.35	1.62
15-20	Mx	4.89	4.48	4.54	6.63
	Mn	-4.24	-4.40	-2.26	-4.18
	Av	1.19	0.60	1.37	0.72
	RMSE	3.04	2.54	2.74	3.35
20-30	Mx	5.94	4.03	7.79	10.44
	Mn	2.51	0.16	4.57	2.96
	Av	4.18	2.38	6.31	5.76
	RMSE	4.33	2.75	6.41	6.20
Overall	Mx	5.94	4.48	7.79	10.44
	Mn	-5.67	-4.46	-5.84	-4.74
	Av	0.17	0.20	0.17	0.59
	RMSE	2.18	1.91	2.84	2.36

the OPELM model can be used to estimate water depth, and this proposed model is a promising method for water depth estimation with a limited number of data points.

Finally, OPELM model results were compared with the IHO standards to assess whether OPELM model accuracy can fulfill hydrographic survey requirements of different orders [4], [44]. The Mx allowable total vertical uncertainty is 1.06 m when survey order 2 (worst case order) is considered at the Av depth of the study area (15 m) using IHO standards S-44 [4], [44]. This indicates that the OPELM model error (RMSE = 1.91 m) cannot meet the requirements of IHO standards for all hydrographic survey orders. However, the developed OPELM model for bathymetric modeling using satellite imagery can be used as a new reliable soft computing technique to reduce shallow-water depth measurement requirements in bathymetric mapping, especially for primary engineering applications when a limited number of "truth" depth data points are used for depth estimation in shallow water.

Satellite-estimated bathymetry using OPELM was used for mapping bathymetry of the study area with 760 930 grid points; the results are presented in Fig. 12(left) and compared with chart depths, Fig. 12(right). The similarities between both maps show promise for using the proposed model at water depths of 20–30 m, even with a small dataset.

From these results, it can be seen that the OPELM can create a reliable bathymetric map even with a small sample set, compared with the state-of-the-art linear regression (Stumpf model), ANN, and PSO-ANN models. Therefore, the OPELM model can enable surveyors to estimate the

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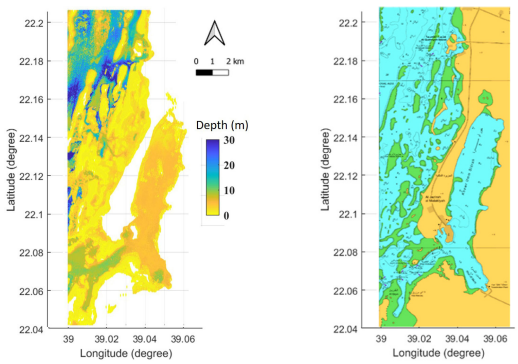


Fig. 12. (Left) Bathymetric map obtained from OPELM model and (Right) nautical chart.

water depth for areas with limited sample data (as presented in Table V), such as in this case study.

IV. Conclusion

This article investigated the development of a satellite-estimated bathymetric model using ANN and hybrid ANN algorithms with a limited number of "truth" depth data points. Two-hybrid ANN algorithms, PSO-ANN and OPELM, were developed and examined based on satellite image data for Alqumriyah Island, Saudi Arabia, along the Red Sea. Sentinel-2 satellite image data and corresponding chart depths were used. Sentinel-2 satellite image bands were adjusted for errors and used for bathymetric modeling. There is a significant relationship between chart depth and input variables of satellite image bands. A comparison was conducted between PSO-ANN, OPELM, Stumpf log ratio, and traditional ANN models to evaluate the proposed models. The performance of the PSO-ANN model is better at the depth range of 0–5 m and the ANN model was the best at detecting the depth range of 5–15 m. The OPELM model was promising for the depth range of 15–30 m. Also, the overall accuracy of OPELM was improved, in terms of RMSE, by 32.46%, 18.76%, and 12.10% compared with Stumpf, ANN, and PSO-ANN models, respectively. The correlations between model results and nautical chart depths were 96.41%, 94.57%, and 95.51% for the training, testing, and all datasets. A good match was also observed when comparing the developed bathymetric map in this article and nautical chart maps, where OPELM shows promising performance for deeper water depth modeling with a limited number of sample data points. Therefore, the developed OPELM model is considered the most accurate model for satellite-derived bathymetry. Finally, OPELM model results were compared with the IHO standards, and the results showed that the OPELM model error does not fulfill the requirements of IHO standards for all hydrographic survey orders. However, the developed OPELM model can be used as a new reliable soft computing technique for bathymetric mapping using satellite imagery, especially for primary engineering applications that are not as stringent in accuracy requirements and that require only a limited number of "truth" depths to develop a model in shallow-water areas. The significant contribution of this research is that it is the first research study investigating the artificial intelligence-based hybrid OPELM method for accurate bathymetric modeling and will certainly lead to further investigations of hybrid models.

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